Saral Singhal

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Education

Vellore Institute of Technology, Bhopal CGPA: 8.42/10
B.Tech., Computer Science and Engineering *2023 – 2027*

Maheshwari Public School, Jaipur, Rajasthan
CBSE, Higher Secondary Education *2016 – 2022*

Technical Skills

Querying & Programming: SQL, Python, Java (Basics)
Analytics & BI: Power BI (DAX, star-schema modelling), Oracle Analytics Cloud, Excel, Matplotlib, Streamlit
Data & ML Libraries: Pandas, NumPy, Scikit-Learn, XGBoost, TensorFlow, PyTorch, Keras
Methods: ETL, Data Modelling, Requirement Gathering, Stakeholder Management
Tools: Git, GitHub, Jupyter Notebook, VS Code, Power Automate, n8n

Experience

Deloitte Touche Tohmatsu India LLP Gurugram, India
Intern, Strategy, Risk & Transactions *May 2026 – August 2026*

- **Feasibility Study:** Contributed to the feasibility study for a 10 MW Tier III data centre in India under a public-private partnership (PPP) model, covering demand assessment, site and infrastructure requirements and a 20-year operating case.
- **Bid Compliance Analytics:** Designed a Power BI dashboard on a star-schema data model with custom DAX measures to score 19 bidders against a 32-point NHAI tender checklist, replacing a manual document-by-document review process.
- **Asset Monetisation Reporting:** Developed the reporting layer for the Asset Registration System, consolidating dispersed highway asset records into review-ready dashboards for stakeholder sign-off.

Ernst & Young LLP Delhi, India
Research Intern *May 2025 – June 2025*

- **Revenue Analysis:** Quantified non-fare box revenue streams for Chennai Metro Rail Limited (CMRL), benchmarking commercial space utilisation and land value capture mechanisms across station assets.
- **Requirements Design:** Specified functional requirements and workflow models for a platform automating vendor-based non-fare revenue collection, tracking and reconciliation.
- **Geospatial Data:** Mapped Chhattisgarh and Uttar Pradesh national highway stretches through GIS, standardising asset records for downstream registration and analysis.

Projects

BleedDetectX | *Python, PyTorch, TensorFlow, Keras* Link

- **Model Development:** Built and compared U-Net, ResNet-UNet and DMP-WCEBleedNet architectures for simultaneous bleeding classification and segmentation in Wireless Capsule Endoscopy imagery.
- **Results:** Reached a Dice coefficient of 0.812 and 98.31% classification accuracy with DMP-WCEBleedNet, outperforming the U-Net baseline on the held-out test set.
- **Evaluation:** Assessed models across Dice, IoU, Precision, Recall and F1 on a class-imbalanced dataset.

TeleTrace | *Python, SQLite, SQL* Link

- **Pipeline Design:** Built a five-stage ETL pipeline in pure Python and SQLite – generate, validate, clean, load and analyse by processing 500 synthetic call-centre records seeded with missing fields, duplicates and malformed timestamps, every rejected record retained a failure reason alongside a run-level ingestion audit log.
- **Modelling & Analysis:** Normalised timestamps to IST and derived duration, hour and weekend attributes with imputation explicitly flagged; loaded into an idempotent SQLite schema so re-runs never duplicated rows, and answered business questions in SQL using conditional aggregation with CSV report exports.

Certifications & Achievements

- Microsoft Certified: Fabric Data Engineer Associate Certificate
- Oracle Analytics Cloud Certified Professional Certificate
- University of MICHIGAN: Applied Machine Learning in Python Certificate
- Published and presented a research paper on Wireless Capsule Endoscopy at ICNACT 2026
- **Co-Lead, Vitronix (Technical Club):** Scripted technical workshops and speaker sessions for a 300+ member community, coordinating design, outreach and logistics teams to ship content on schedule and support 110+ registrations..
- **Co-Lead, Mahro Rajasthan Club (Cultural Club):** Led a 7-member team across 4 cultural events attended by 500+ students; standardised a bilingual script template and two-round review process that cut last-minute rewrites.
- **Sport & Interests:** Represented college clubs in badminton; keen reader of mystery and crime fiction.